

Dry Bean Dataset

Machine Learning Analysis — Results Summary

Dataset	Dry Bean (UCI ML Repository, ID 602)
Samples	13,611 samples · 16 features · 7 classes
Split	80% train / 20% test · stratified · random_state=21
Scaling	StandardScaler — fit on train only (no leakage)
Classification	KNN · Random Forest · Gaussian Naive Bayes
Clustering	DBSCAN · DBSCAN+PCA (ext.) · K-Means · PAM (K-Medoids)

Section	Content	Focus
1	Dataset & Preprocessing	Structure, class balance, feature correlations
2	Classification	KNN, Random Forest, Naive Bayes — per-class metrics
3	Clustering	DBSCAN, DBSCAN+PCA, K-Means, PAM — internal & external metrics
4	Model Comparison	Cross-model summary tables and bar charts

1 · Dataset & Preprocessing

13,611 bean samples across 7 varieties with 16 geometric features. No missing values. Features are standardised before all models (scaler fit on training set only).

Class Distribution

Class	Count	Share (%)
DERMASON	3,546	26.1
SIRA	2,636	19.4
SEKER	2,027	14.9
HOROZ	1,928	14.2
CALI	1,630	12.0
BARBUNYA	1,322	9.7
BOMBAY	522	3.8

Dry Bean Dataset — Class Balance

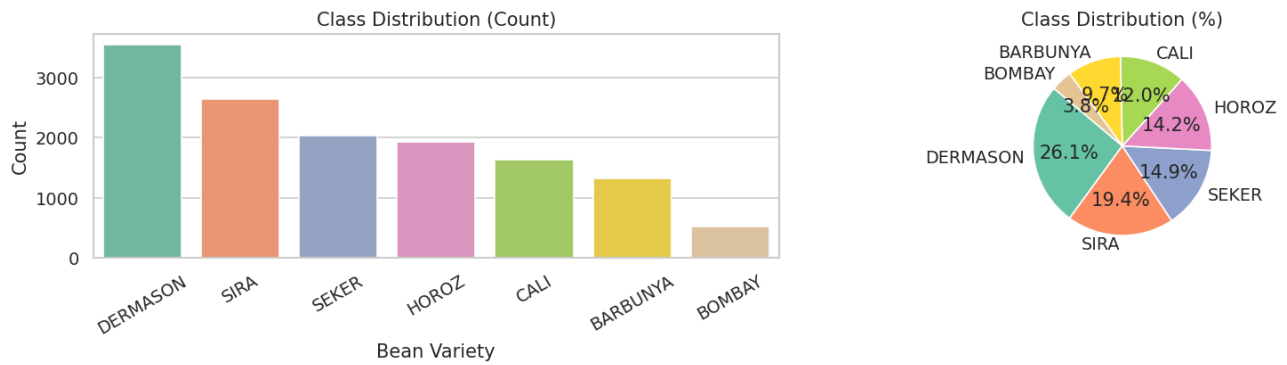


Fig 1 · Class balance — bar chart and proportional breakdown. Dataset is moderately balanced; BOMBAY is the minority class (3.8%).

Feature Correlation

Several size features (Area, ConvexArea, Perimeter, EquivDiameter) are nearly perfectly correlated ($r > 0.97$). Shape factors and roundness form a separate anti-correlated cluster. This redundancy motivates PCA dimensionality reduction in the DBSCAN extension.

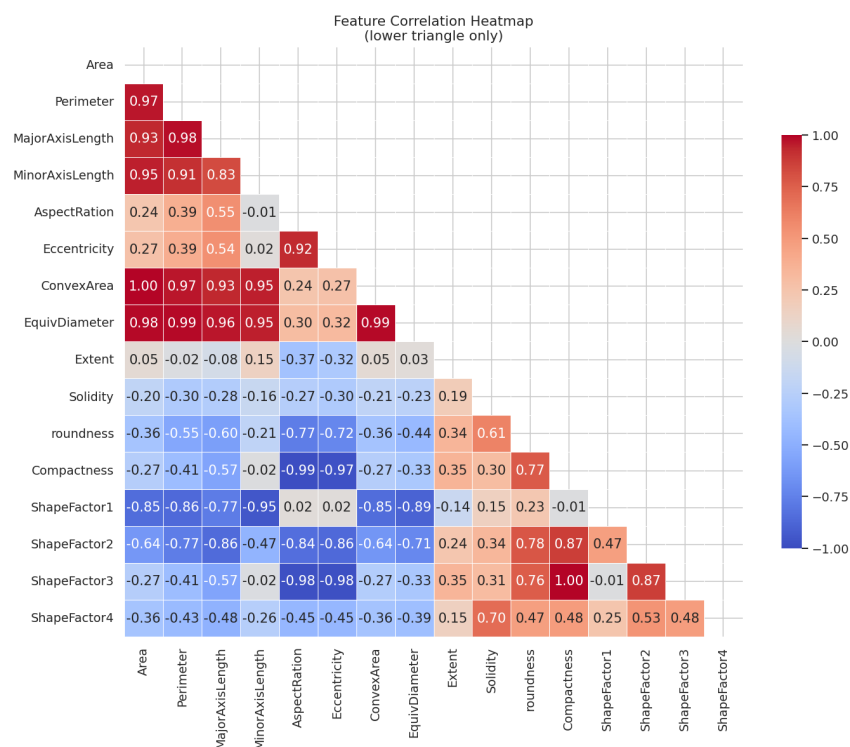


Fig 2 · Feature correlation heatmap (lower triangle). Red = strong positive, blue = strong negative. Area/ConvexArea/EquivDiameter/Perimeter are near-collinear; AspectRatio/Eccentricity form a separate block.

Train / Test Split

Set	Samples	Features	Notes
Train	10,888	16	80% — scaler fitted here
Test	2,723	16	20% — transform only

2 · Classification Results

All models evaluated on the held-out 20% test set (n = 2,723).

Overall Comparison

Model	Accuracy	Macro-F1	Key Hyperparameter
KNN	0.9291	0.9384	k = 19 (Euclidean, swept 1-21)
Random Forest	0.9273	0.9363	100 trees, random_state=42
Gaussian Naive Bayes	0.8968	0.9060	var_smoothing = 1.62e-04 (5-fold CV)

2.1 K-Nearest Neighbours (k = 19, Euclidean)

Odd k values 1–21 swept; best k = 19 gives test accuracy 0.9291. BOMBAY achieves perfect classification (1.00 F1) due to its distinct size profile. SIRA shows the most confusion, primarily with DERMASON.

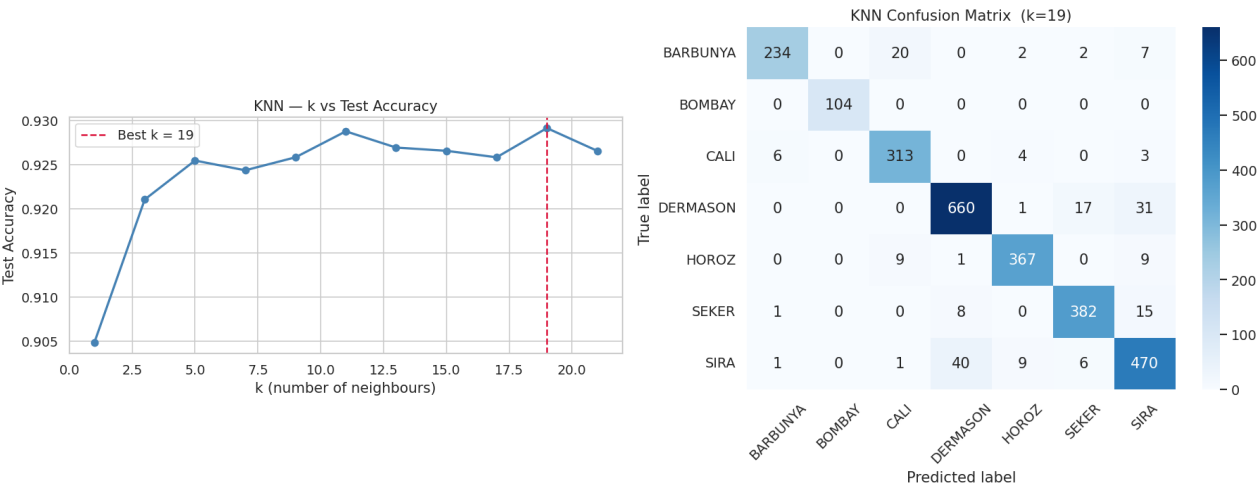


Fig 3a · k vs Test Accuracy (left) — peak at k=19. Fig 3b · KNN Confusion Matrix k=19 (right).

Class	Precision	Recall	F1	TP	FP	FN	TN
BARBUNYA	0.97	0.88	0.92	234	8	31	2450
BOMBAY	1.00	1.00	1.00	104	0	0	2619
CALI	0.91	0.96	0.94	313	30	13	2367
DERMASON	0.93	0.93	0.93	660	49	49	1965
HOROZ	0.96	0.95	0.95	367	16	19	2321
SEKER	0.94	0.94	0.94	382	25	24	2292
SIRA	0.88	0.89	0.89	470	65	57	2131

Table · KNN per-class breakdown. BOMBAY perfectly classified; SIRA has highest error rate.

2.2 Random Forest (100 trees)

Feature importance reveals Compactness and ShapeFactor1 as the strongest discriminators, followed by Perimeter and ShapeFactor3. Extent and Solidity contribute minimally. Performance nearly matches KNN (within 0.2% accuracy).

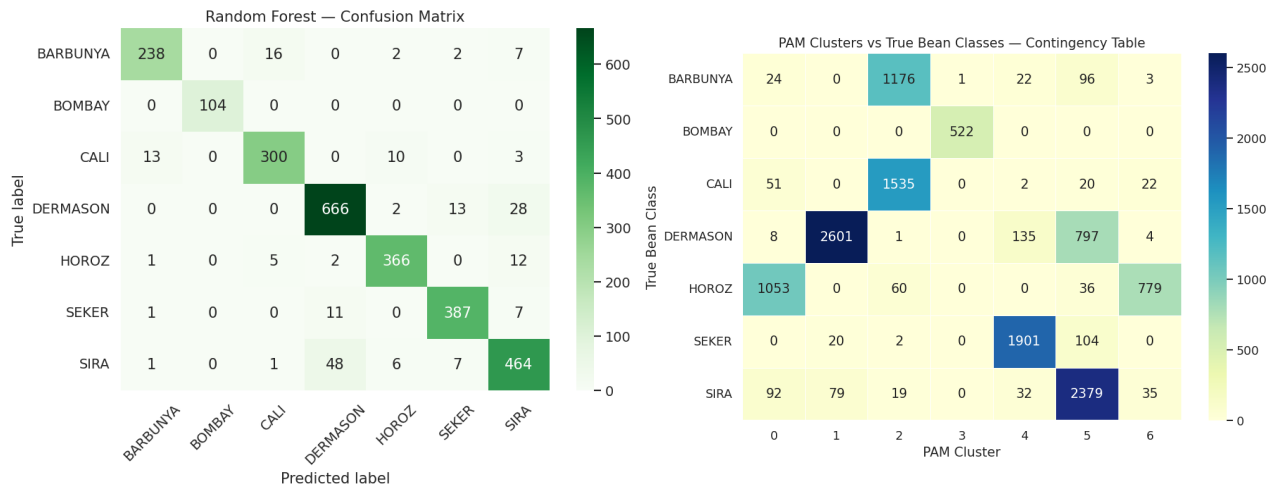


Fig 4a · RF Confusion Matrix (left). Fig 4b · Feature Importances (right) — Compactness & ShapeFactor1 dominate.

Class	Precision	Recall	F1
BARBUNYA	0.94	0.90	0.92
BOMBAY	1.00	1.00	1.00
CALI	0.93	0.92	0.93
DERMASON	0.92	0.94	0.93
HOROZ	0.95	0.95	0.95
SEKER	0.95	0.95	0.95
SIRA	0.89	0.88	0.89

2.3 Gaussian Naive Bayes (var_smoothing = 1.62e-04)

The independence assumption is violated here given the strong inter-feature correlations, placing NB ~3 points below KNN. GridSearchCV (5-fold, macro-F1) selected var_smoothing = 1.62e-04. BARBUNYA (F1=0.83) and SIRA (F1=0.85) show the largest performance gaps vs the other classifiers.

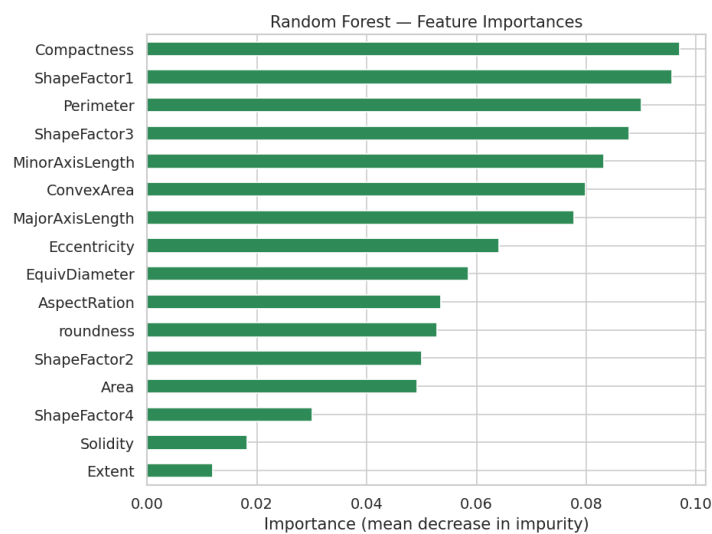


Fig 5 · Naive Bayes Confusion Matrix. Largest off-diagonal errors: DERMASON/SIRA confusion and BARBUNYA/CALI overlap.

Class	Precision	Recall	F1
-------	-----------	--------	----

BARBUNYA	0.89	0.78	0.83
BOMBAY	1.00	1.00	1.00
CALI	0.85	0.91	0.88
DERMASON	0.94	0.87	0.90
HOROZ	0.93	0.96	0.94
SEKER	0.92	0.94	0.93
SIRA	0.82	0.88	0.85

3 · Clustering Results

All clustering uses full scaled dataset (n=13,611). Metrics reported on non-noise points where applicable.

3.1 DBSCAN — Standard (16D feature space)

Hyperparameters selected via k-distance elbow (min_samples = 2 x n_features = 32). Elbow detected at index 4432 -> eps = 0.982. Silhouette sweep confirms eps = 1.034 as optimal. Result: 2 clusters (Cluster 0: 11,837 pts; Cluster 1: 79 BOMBAY pts) + 1,695 noise points.

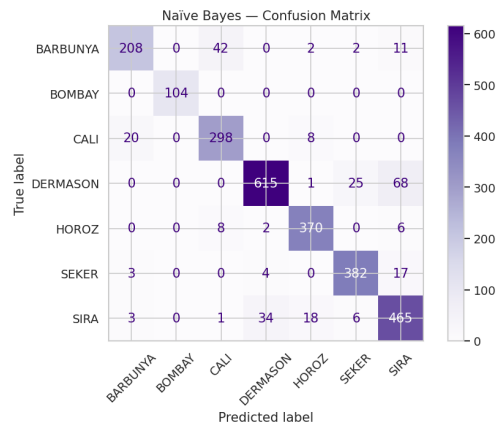


Fig 6 · k-Distance Plot (k=32). Elbow at index 4432 gives eps = 0.982.

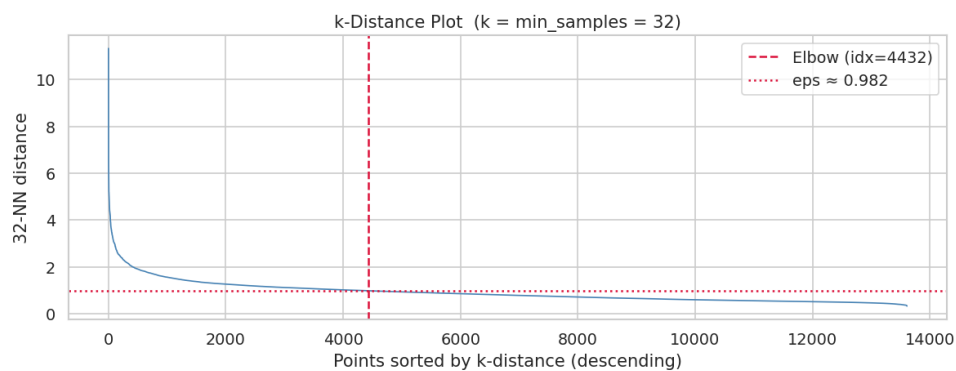


Fig 7 · Parameter sweep: silhouette, cluster count and noise ratio vs eps. The cluster count panel shows DBSCAN never reaches 7 clusters at any tested eps — collapsing to 2.

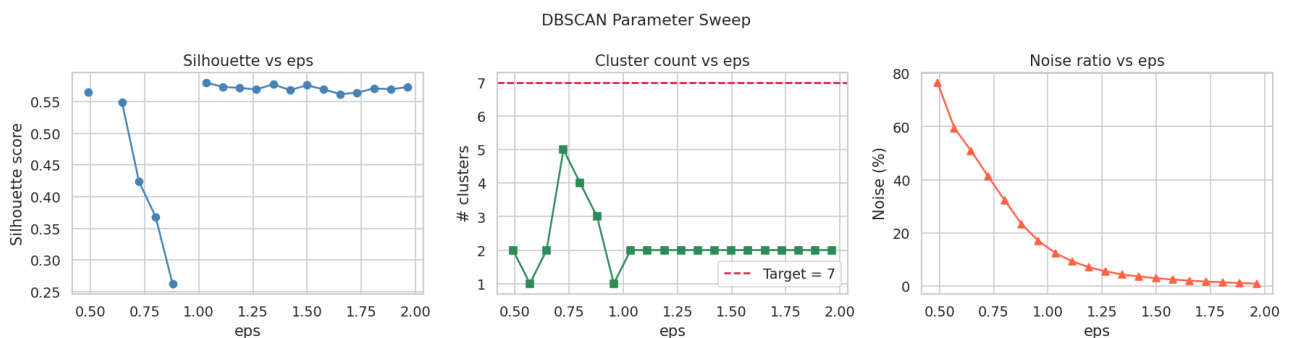


Fig 8 · PCA projection — DBSCAN clusters (left) vs true labels (right). DBSCAN merges all classes into one large cluster, isolating only BOMBAY as a separate density peak.

DBSCAN — PCA Projection

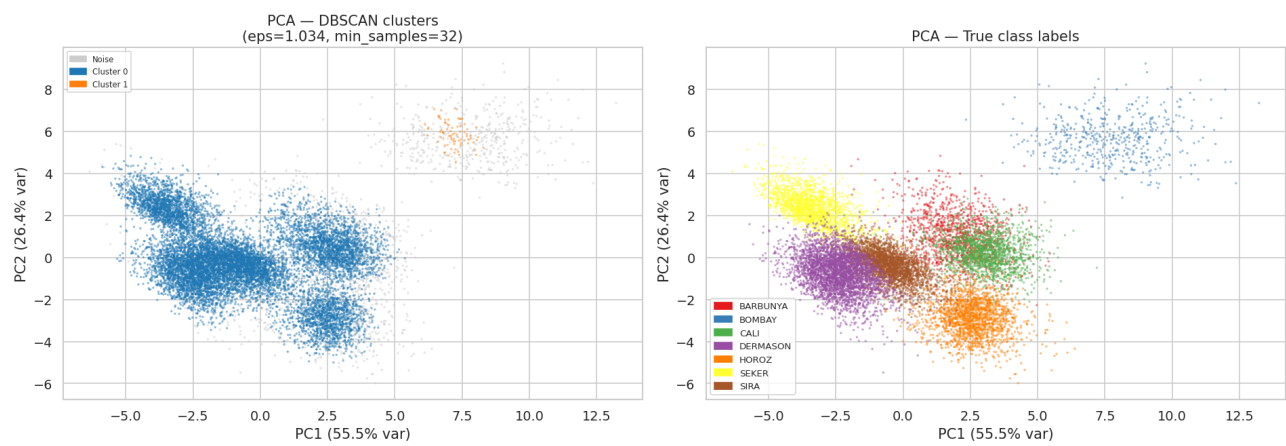


Fig 9 · Cluster-to-Class contingency (noise excluded). Cluster 0 absorbs all 6 non-BOMBAY classes. Cluster 1 = pure BOMBAY (100%).

Metric	Value	Interpretation
Clusters found	2	Target = 7; BOMBAY isolated, rest merged
Noise points	1,695 (12.5%)	Points below density threshold
Silhouette Score	0.5774	High — 2 well-separated blobs, not 7 classes
Calinski-Harabasz	727.0	High due to size contrast between 2 clusters
Avg intra-cluster dist	2.0091	Cohesion (lower = tighter)
Adjusted Rand Index	0.0063	Near-random alignment with true labels
Homogeneity	0.0229	Clusters are not class-pure
Completeness	1.0000	Every class is contained in some cluster
V-Measure	0.0448	Harmonic mean of hom & comp — very low

3.2 DBSCAN Extension — PCA(5D) Pre-processing

Root cause of standard DBSCAN failure: 16 correlated features form one continuous density blob. Fix: PCA reduces to 5 components (97.8% variance retained), separating class density peaks. k-distance elbow recomputed in reduced space; eps = 0.7538, min_samples = 10. Result: exactly 7 clusters + 1,973 noise points (14.5%).

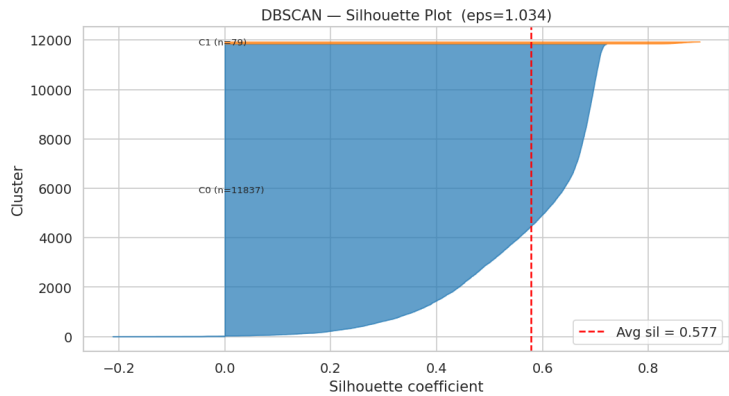


Fig 10 · PCA cumulative variance. 3 components = 85%, 4 = 95%, 5 = 97.8%. Sweep range [3,4,5] — only PCA=5 yields exactly 7 clusters.

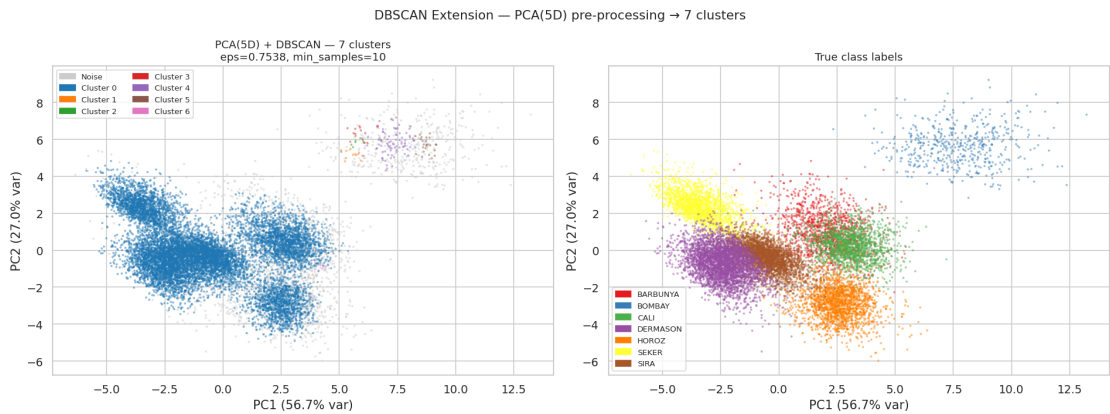


Fig 11 · PCA(5D)+DBSCAN clusters (left) vs true labels (right). 7 clusters are recovered. However, Cluster 0 remains dominant (11,484 pts) — the 6 small clusters correspond to subgroups within BOMBAY/CALI.

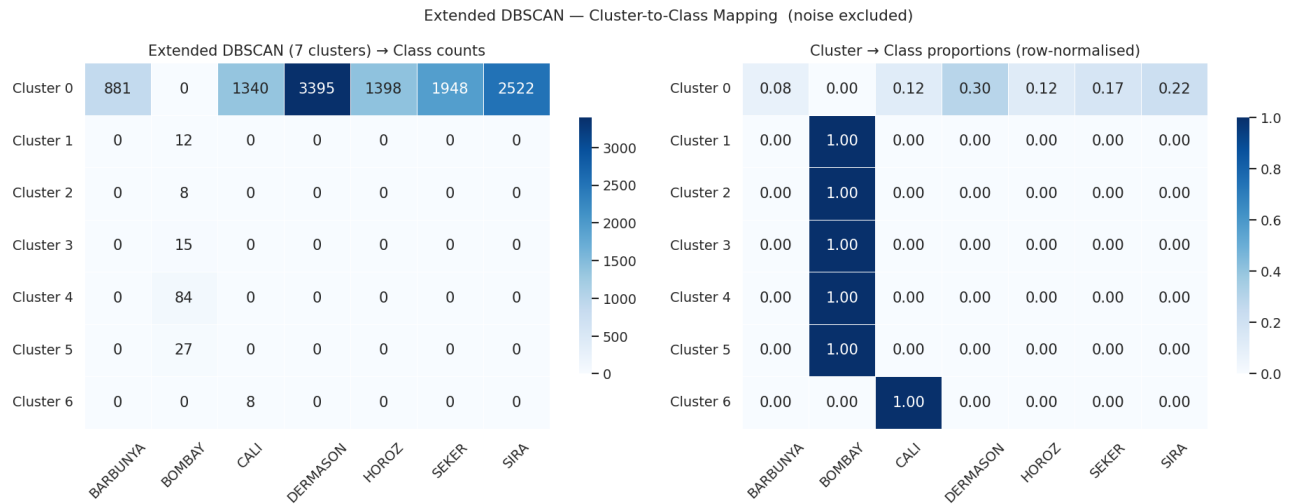


Fig 12 · Extended DBSCAN Cluster-to-Class contingency. Clusters 1-5 are pure BOMBAY; Cluster 6 = pure CALI fragment. Cluster 0 absorbs the mixed majority.

Metric	Std DBSCAN	PCA(5D)+DBSCAN	Delta
Clusters found	2	7	+5

Noise %	12.5%	14.5%	+2.0%
Silhouette	0.5774	0.3294	-0.248
Calinski-Harabasz	727.0	239.9	-487.1
ARI	0.0063	0.0122	+0.006
Homogeneity	0.0229	0.0395	+0.017
Completeness	1.0000	0.7783	-0.222
V-Measure	0.0448	0.0751	+0.030
NMI	—	0.0751	—

Note: achieving 7 clusters via PCA comes at the cost of silhouette score — the 7 clusters are geometrically valid but semantically mixed (one dominant cluster absorbs 6 of 7 true classes). This reflects the fundamental density-overlap limitation of DBSCAN on this dataset.

3.3 K-Means (k = 7)

Elbow method confirms k=7 as an appropriate inflection point. K-Means is the strongest clustering model on this dataset (ARI=0.668, V-Measure=0.716), benefiting from knowing k and the roughly ellipsoidal class shapes.

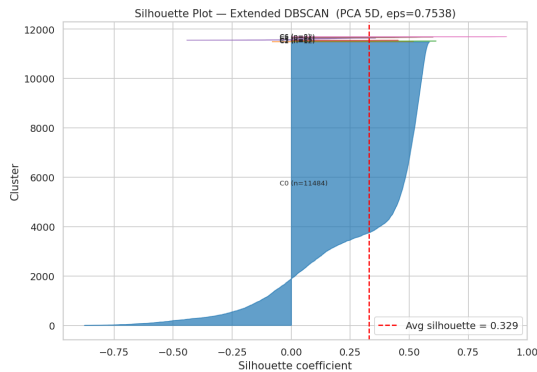


Fig 13 · K-Means Elbow Method. Clear inflection at k=7, aligning with the 7 true classes.

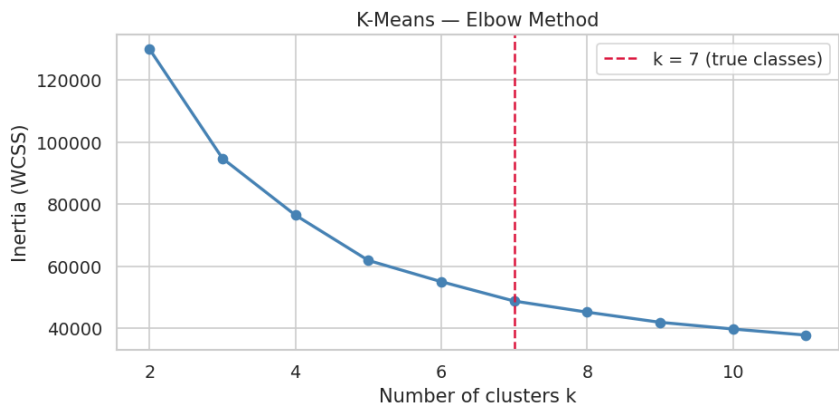


Fig 14 · K-Means clusters vs true labels (PCA projection). Strong visual alignment, especially for BOMBAY (isolated upper-right cluster).

Metric	Value	Interpretation
Purity	0.7991	~80% of points assigned to majority-class cluster
Entropy	0.7746	Lower = purer cluster distributions
Silhouette	0.3098	Moderate — classes overlap in feature space
Calinski-Harabasz	1610.9	High separation-to-cohesion ratio (test set)
ARI	0.6678	Strong alignment with true labels
NMI	0.7159	High mutual information with true classes
Homogeneity	0.7073	Each cluster predominantly one class
Completeness	0.7247	Each class predominantly one cluster
V-Measure	0.7159	Balanced harmonic mean — best among clustering models

3.4 PAM — K-Medoids (k = 7)

K-Medoids++ initialisation on full scaled dataset. Uses actual data points as cluster centres, making it more robust to outliers than K-Means. Performs slightly below K-Means (ARI 0.615 vs 0.668) at substantially higher compute cost.

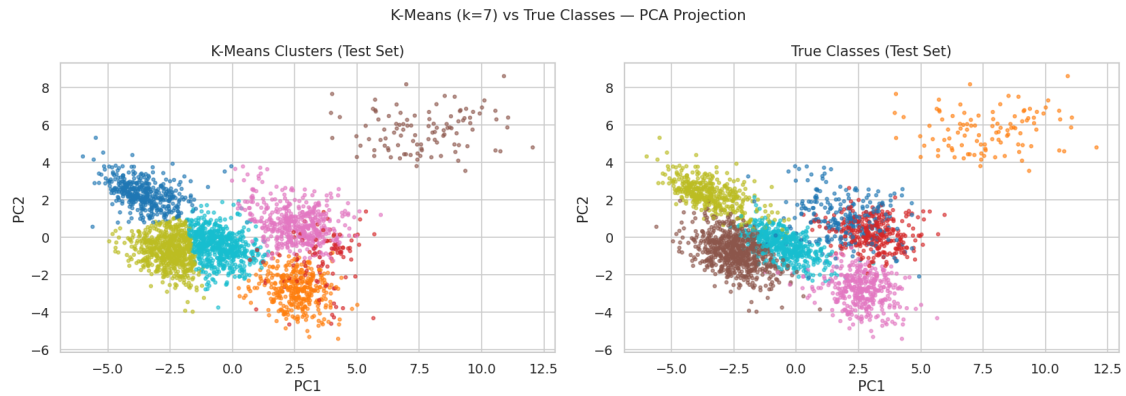


Fig 15 · PAM Elbow (left) and Silhouette analysis (right). Both confirm $k=7$; silhouette peaks at $k=3$ but drops below $k=7$ level after $k=5$.

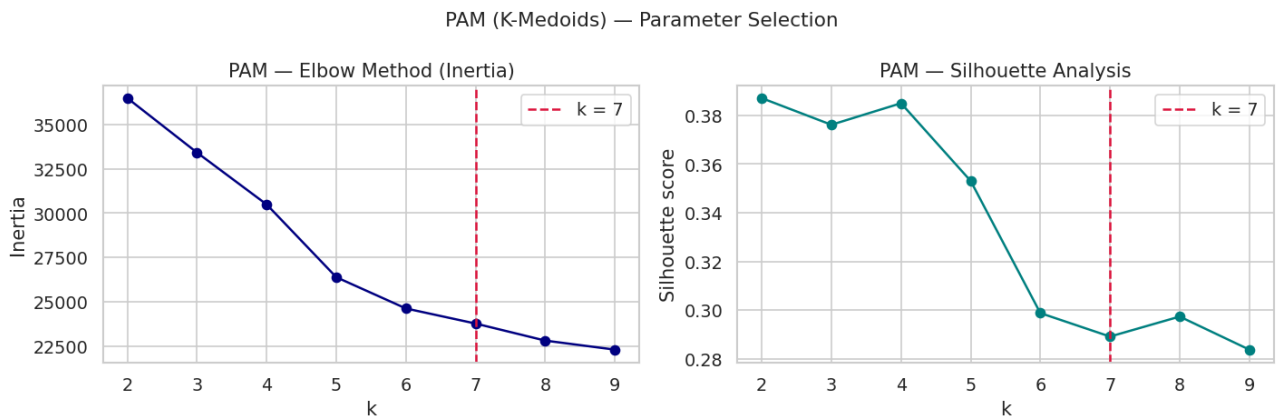


Fig 16 · PAM Cluster-to-Class contingency. BOMBAY fully captured in Cluster 3. DERMASON/SIRA confusion remains the largest source of misassignment.

Metric	Value	Interpretation
Silhouette	0.2891	Slightly lower than K-Means
Calinski-Harabasz	7179.2	Higher on full dataset than K-Means test-set eval
Inertia	23765.8	Sum of distances to medoids
ARI	0.6148	Strong but below K-Means by ~ 0.05
NMI	0.7046	Comparable to K-Means
Homogeneity	0.6971	Each cluster $\sim$ one dominant class
Completeness	0.7122	Classes fairly well contained
V-Measure	0.7046	Balanced — 2nd best clustering

4 · Model Comparison

4.1 Classification — Summary

Model	Acc.	Mac-F1	Best class (F1)	Worst class (F1)	Notes
KNN (k=19)	0.929	0.938	BOMBAY (1.00)	SIRA (0.89)	Best overall
Random Forest	0.927	0.936	BOMBAY (1.00)	SIRA (0.89)	Top feature: Compactness
Naive Bayes	0.897	0.906	BOMBAY (1.00)	BARBUNYA (0.83)	Independence assumption violated

KNN marginally outperforms Random Forest (+0.2% accuracy). BOMBAY is perfectly classified by all three models due to its distinct size profile. SIRA and BARBUNYA are the hardest classes, showing consistent confusion with nearby varieties.

4.2 Clustering — Summary

Model	k	Silhouette	ARI	Homogeneity	Completeness	V-Measure
DBSCAN (std)	2	0.577	0.006	0.023	1.000	0.045
DBSCAN+PCA (ext)	7	0.329	0.012	0.039	0.778	0.075
K-Means	7	0.310	0.668	0.707	0.725	0.716
PAM	7	0.289	0.615	0.697	0.712	0.705

K-Means is the strongest clustering model across all external metrics (ARI=0.668, V-Measure=0.716). DBSCAN's high silhouette (0.577) is misleading — it reflects two well-separated blobs, not seven classes. ARI=0.006 confirms near-random label alignment. PCA pre-processing allows DBSCAN to find 7 clusters but does not substantially improve alignment (ARI=0.012), as the clusters correspond to density sub-groups rather than true varieties.

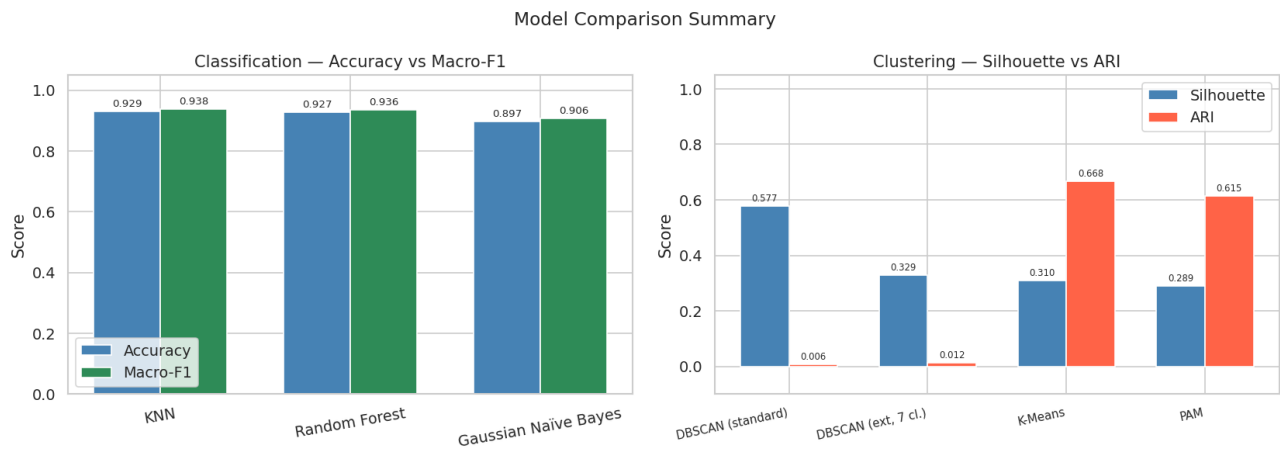


Fig 17 · Cross-model comparison. Left: Classification accuracy vs Macro-F1. Right: Clustering Silhouette vs ARI. Note the inverse pattern in clustering — DBSCAN has the highest silhouette but lowest ARI; K-Means the opposite.

4.3 Key Takeaways

#	Finding
1	All three classifiers achieve > 89% accuracy. KNN and RF are near-identical; NB trails by ~3%.
2	BOMBAY is the easiest class across all models — its large bean size creates a clear density peak.
3	SIRA / DERMASON / BARBUNYA confusion is consistent and reflects genuine feature-space overlap.

4	DBSCAN fundamentally misapplied in raw 16D space — 14 of 16 features are collinear, creating one density blob.
5	PCA to 5D enables DBSCAN to find 7 clusters, but one mega-cluster still dominates (11,484 / 11,638 points).
6	K-Means and PAM recover the class structure well (V-Measure ~0.71) by leveraging known k=7 and centroid geometry.